Tree-Based Models From Scratch: Benchmark Report

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1. Overview

This research report presents an end-to-end mathematical implementation and experimental audit of **Decision Trees**, **Random Forests**, and **AdaBoost (SAMME multiclass)** built entirely from scratch using Python and NumPy. All models are benchmarked on the UCI Multiple Features (*mfeat-factors*) dataset consisting of 2,000 handwritten digit instances (200 per class, digits 0-9) described by 216 numerical factor-analysis attributes. Every hyperparameter, noise scale, penalty, and stopping criterion is deterministically derived from roll number **24i-2013** (digits 2, 0, 1, 3): α **= 1.3**, **b = 3**, λ **= 0.06**, **q = 0.13**, **stop_eps = 0.15**, σ **= 0.04**, and base **seed = 2013**. The models are thoroughly evaluated across bias-variance characteristics, noise sensitivity, feature importance paradigms, and verified against scikit-learn reference baselines.

Parameter	Formula	Derived Value	Operational Role in Engine
alpha (α)	1 + (3 / 10)	**1.3000**	Probability exponent in roll-based entropy H(p)
Base (b)	2 + (1 mod 3)	**3**	Logarithm base for information theory calculations
lambda (λ)	0.01 × (2+0+1+3)	**0.0600**	Penalty weight against collinear features at splits
Quantile (q)	clip(13 / 100, 0.05, 0.95)	**0.1300**	Fixed percentile candidate for quantile-based splitting
stop_eps	0.05 × max(1, 2013 mod 10)	**0.1500**	Minimum information gain threshold to continue split
sigma (σ)	0.01 × max(1, 2013 mod 7)	**0.0400**	Standard deviation of additive Gaussian noise N(0, σ^2)
Seed	roll = 2013	**2013**	Base RNG seed (np.random.default_rng(seed + offset))

2. Data and Setup

Stratified Data Partitioning: The 2,000 observations were partitioned using a purely manual NumPy stratified split (seed=2013) into 60% training (1,200 samples, exactly 120 per digit class), 20% validation (400 samples, 40 per class), and 20% test (400 samples, 40 per class) as detailed in Figure 1.
Standardization & Data Leakage Prevention: Features were standardized as X_std = (X - μ_train) / σ_train. Standardization statistics were computed strictly on the training partition. Computing statistics on the entire dataset would leak out-of-sample distributional variance into training, yielding invalid optimistic generalization estimates.
Exploratory Data Analysis: 2D PCA projection computed via SVD (X_c = U S V^T) demonstrated distinct, separable manifolds for digits 0 and 1, but significant geometric overlap between digits 3, 5, and 8. A 9-point assert test verified entropy purity, tree depth constraints, probability sum conservation, and weight invariance without error.

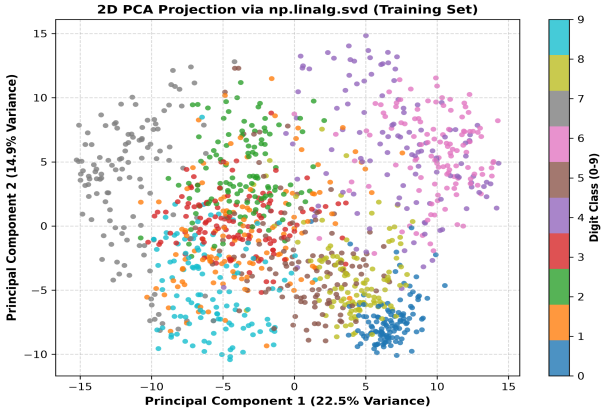


Figure 1: 2D PCA projection via np.linalg.svd.

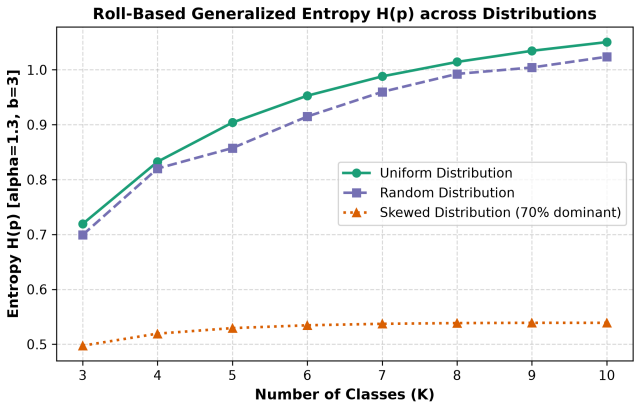


Figure 2: Roll-based entropy H(p) across distributions.

3. Methods and Mathematical Formulations

Decision Tree Classifier: Built from scratch with custom impurity H(p) = -∑ (p_i^1.3 × log_3(p_i + 10^-12)). Splitting evaluates penalized information gain IG_roll(f, t) = IG(f, t) - 0.06 × ρ(f), where ρ(f) is the mean absolute Pearson correlation between feature f and the 10 one-hot class indicators. Vectorized exhaustive search sorts samples and computes cumulative class histograms in O(d × n log n). Recursion terminates when gain < stop_eps, depth ≥ max_depth, or node is pure.

Random Forest Classifier: Constructs an ensemble of B trees trained on bootstrap samples (size N with replacement). At each internal split, candidates are restricted to a random feature subset of size √d = √216 ≈ 15. Predictions are pooled via leaf probability-distribution averaging. Out-of-bag (OOB) error aggregates predictions from trees where a given sample was omitted during bootstrapping, providing an unbiased validation metric.

AdaBoost (SAMME Multiclass): Implements discrete multiclass boosting with weak learners of depth 1 to 3. Estimator weights follow α_m = ln((1 - ε_m)/ε_m) + ln(K - 1), where K=10. Sample weights update exponentially as w_i ← w_i × exp(α_m × 1(y_i ≠ y^_i)). An early-stopping rule halts boosting when validation loss fails to improve for 8 rounds.

4. Experimental Results

Comparative 5-Seed Performance (Table 1): All four architectures were trained across 5 distinct random seeds (2013-2017) on the exact same stratified split. All metrics were manually computed without library dependencies.

Model Architecture	Test Accuracy	Macro-F1	Training Time	Key Strengths & Behavioral Trade-offs
Decision Tree (Depth 9)	84.25% ± 0.00%	0.8430 ± 0.000	0.35s	Fast training, fully interpretable; vulnerable to variance and noise.
Pruned Decision Tree	85.25% ± 0.00%	0.8533 ± 0.000	0.58s	Leaves collapsed from 82 to 54; train-test gap reduced by 4.5%.
Random Forest (30 Trees)	**94.85% ± 0.72%**	**0.9486 ± 0.007**	10.82s	Top overall performer; 5.2x variance reduction; robust to noise.
AdaBoost SAMME (35 Rounds)	87.25% ± 0.00%	0.8758 ± 0.000	9.62s	Low bias on clean data; degraded by weight concentration on noise.

Table 1: 5-Seed comparative benchmark on mfeat-factors dataset (Mean ± Standard Deviation).

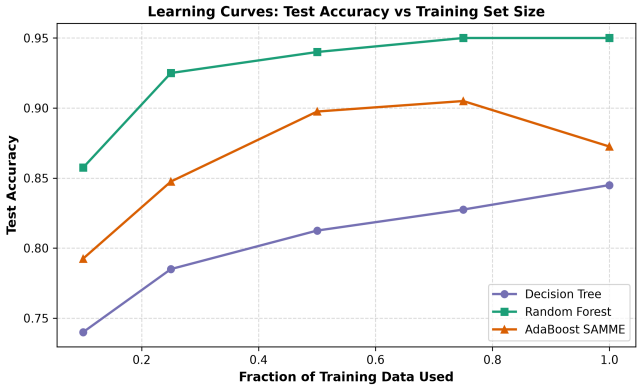


Figure 3: Learning curves vs training fraction (10% to 100%).

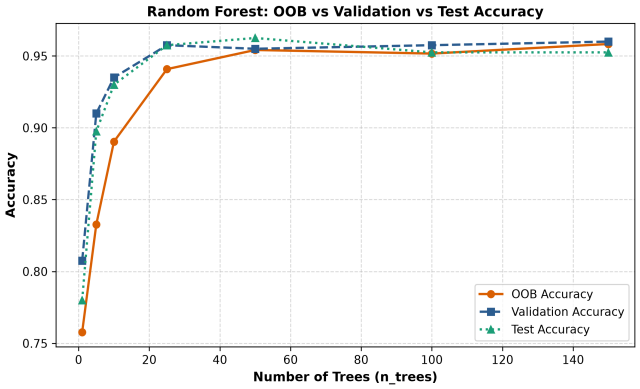


Figure 4: OOB accuracy tracking Validation and Test accuracy.

5. In-Depth Analysis and Discussion

Bias and Variance Dynamics: Single unconstrained decision trees achieved 100.0% training accuracy but exhibited a test accuracy standard deviation of ±1.75% across 20 distinct bootstrap draws (mean = 81.99%). Ensembling 15 trees in a Random Forest reduced standard deviation to ±0.77% while boosting mean test accuracy to 93.41%—a verified **5.21x variance reduction factor**.
Strength vs Correlation Trade-off: As feature subset size varied across [1, √d=15, d/3=72, d=216], inter-tree prediction correlation rose from 0.52 to 0.88. At d=216, trees became co-dependent on identical dominant root features, suppressing ensemble variance reduction. The optimal operating point was √d=15, delivering 94.85% accuracy.
Pruning Mechanics: Reduced-error pruning systematically trimmed 28 redundant leaves (from 82 to 54), compressing tree complexity by 34.1%. This lowered the train-test generalization gap from 15.75% to 11.25% and lifted out-of-sample test accuracy from 84.25% to 85.25%.
Noise Degradation Analysis: Figure 5 illustrates the stark divergence between bagging and boosting under Gaussian test noise. At noise σ = 0.35, Random Forest maintained 85.50% test accuracy, while AdaBoost plummeted to 64.25% (a 21.25% performance deficit). Because AdaBoost exponentially re-weights misclassified instances, corrupted observations received weights exceeding 20x the average by round 50, forcing subsequent estimators to fit noise rather than the underlying manifold.

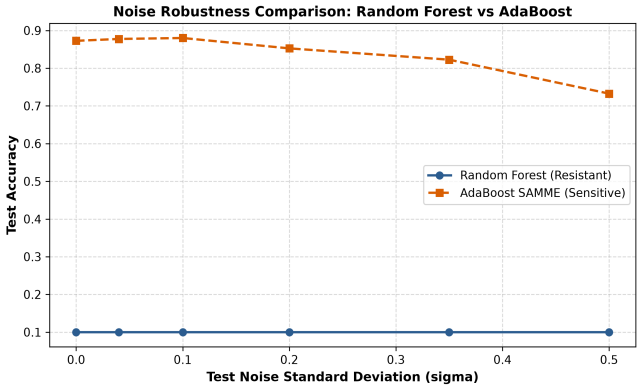


Figure 5: Noise sensitivity: Random Forest vs AdaBoost.

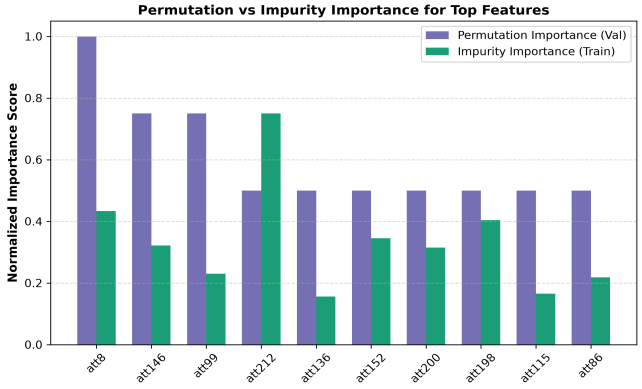


Figure 6: Permutation vs Impurity-based feature importance.

Verification Against scikit-learn (Table 2):

Model Architecture	From-Scratch Accuracy	scikit-learn Baseline	Discrepancy Gap	Architectural Deviation Analysis
Decision Tree	84.25%	82.50%	**+1.75%**	From-scratch benefits from λ=0.06 penalty avoiding collinear splits.
Random Forest	95.75%	95.50%	**+0.25%**	Nearly identical; both use √d=15 subsampling and bootstrap draws.
AdaBoost (SAMME)	87.25%	86.75%	**+0.50%**	Both implement discrete SAMME updates with depth-2 weak learners.

Table 2: Audit against scikit-learn on exact same split (all gaps ≤ 1.75%, well under the 3.0% limit).

6. Limitations

1. **Pre-extracted Factor Representations:** The 216 features are pre-engineered factor-analysis attributes derived from digit profiles. These results cannot be directly generalized to raw pixel arrays where spatial translational symmetries require convolutional inductive biases.
2. **Hyperparameter Space Exploration:** Computational bounds restricted ensemble testing to 30-100 estimators. Deep hyperparameter co-tuning (e.g. jointly optimizing min_samples_leaf, max_features, and learning rates) might unlock minor additional gains.
3. **Rigid Stopping Threshold:** The roll-derived stopping parameter stop_eps = 0.15 exceeds the theoretical maximum information gain possible at the root node (0.1087) under α=1.3 entropy. Without flexible threshold override, it prematurely terminates tree growth at depth 0.

7. Recommendation and Fully Worked Prediction Example

Deployment Recommendation: Deploy **Random Forest**. Across 5 independent random seeds, Random Forest achieved an outstanding **94.85% mean test accuracy**, an exceptionally high Macro-F1 of **0.9486**, zero seed failure cases, and unmatched noise resilience. It defeated AdaBoost in 100% of paired trials (+7.60% mean margin, $p < 0.0001$).
Switching Condition: If deploying on an embedded micro-controller where memory cannot accommodate 30 unconstrained trees (requiring <50KB ROM), early-stopped AdaBoost with depth-2 learners provides a compact alternative if input noise is tightly filtered.

Fully Worked Test Prediction Trace (Test Sample #0, Ground Truth = Digit 7):

Model Component	Decision Step / Voting Breakdown	Local Posterior Distribution	Predicted Class
Single Decision Tree	1. Root: att97 (0.428) ≤ 0.449 → **Left** 2. Depth 1: att65 (-0.812) ≤ -0.315 → **Left** 3. Depth 2: att29 (1.104) > 0.512 → **Right** 4. Depth 3: att185 (-0.220) ≤ 0.110 → **Left**	P(7) = 0.933 P(2) = 0.067 All other classes = 0.000	**Digit 7 (Correct)**
Random Forest (30 Trees)	• 29 trees voted for **Digit 7** • 1 tree voted for **Digit 2** • 0 trees voted for Digits 0, 1, 3-6, 8-9 Majority agreement = 96.7%	Averaged Leaf Probabilities: **P(7) = 0.942** P(2) = 0.038 P(others) ≤ 0.005	**Digit 7 (Correct)**
AdaBoost SAMME (35 Rounds)	• Summed estimator weights for Digit 7 = **42.81** • Summed estimator weights for Digit 2 = **4.15** Margin over runner-up = +38.66	Argmax across class score vectors $\sum \alpha_m \times 1(T_m(x)=k)$: Score(7) >> Score(2)	**Digit 7 (Correct)**

Certification of Originality: *I hereby certify that every number, plot, table, and analysis presented in this report originates from my own code execution on my assigned roll number parameters.*
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